# THE EXT^1-ANNIHILATED PROPERTY PASSES ALONG MORPHISMS ISOMORPHIC IN DEGREE ZERO AND SURJECTIVE IN DEGREE -1

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Ext^0_R(M[1], N) = Ext^{-1}_R(M, N) vanishes for a module M and non-positive cohomological degree. invoked at proof lines 22624-22625. Makes the connecting map zero so Ext^1(K', N) injects into Ext^1(K, N).

*Source reference.* Original lemma 0G9F: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22452–22452. Cited at statement lines 22606-22607 for the property being transferred; the equivalence is used inside the counted span.

**Lemma 0.2.** *Let $I$ be an ideal of a ring $R$. Let $\alpha : K \to K'$ be a morphism of $D(R)$. Assume*

(1) *$H^i(K) = H^i(K') = 0$ for $i \notin \{-1, 0\}$*
(2) *$H^0(\alpha)$ is an isomorphism and $H^{-1}(\alpha)$ is surjective.*

*If $K$ satisfies the equivalent conditions (1), (2), and (3) of Lemma 0G9F, then $K'$ does too.*

*Proof.* Set $M = \mathrm{Ker}(H^{-1}(\alpha))$. Then $\alpha$ fits into a distinguished triangle

$$M[1] \to K \to K' \to M[2]$$

For any $R$-module $N$ this determines an exact sequence

$$\mathrm{Ext}^0_R(M[1], N) \to \mathrm{Ext}^1_R(K', N) \to \mathrm{Ext}^1_R(K, N)$$

Since $\mathrm{Ext}^0_R(M[1], N) = \mathrm{Ext}^{-1}_R(M, N) = 0$ we see that $\mathrm{Ext}^1_R(K', N)$ is a submodule of $\mathrm{Ext}^1_R(K, N)$. Hence if $\mathrm{Ext}^1_R(K, N)$ is annihilated by $I$ so is $\mathrm{Ext}^1_R(K', N)$. □

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.